PROJECT TITLE :

"Toy Shop Management System: A Seller and Buyer Program"

PROBLEM DESCRIPTION / OVERVIEW :

This program is a command-line-based toy store system where users can act as either a Seller or a Buyer.

- The Seller can add toys, set prices, and manage stock.
- The Buyer can view available toys and purchase them using a balance.
- The system uses a dictionary (I) to store product data such as price and stock.
- The program runs in a loop until the user chooses to exit.

It simulates a simple inventory management + shopping system.

HOW TO RUN THE PROGRAM :

1. Make sure you have Python installed (Python 3.x)
2. Copy the code into a .py file. example: toy_store.py
3. Run the program using: python toy_store.py
4. Follow the on-screen menu:
   Choose Seller (a) → `manage inventory`
   Choose Buyer (b) → `buy items`
   Choose Exit (c) → `stop program`

SAMPLE INTERACTION / EXAMPLE OUTPUT :

```
Start:
 Pick a Role:
 a. Seller
 b. Buyer
 c. Exit
 Input: a

Seller adds product:
 ----[TOY MAKER]----
 Enter a toy name: Teddy
 Enter price: 100
 Enter quantity: 5

[----Added Teddy to inventory]----
```

Buyer view:
----[SHOP]----
Balance: 1000

----[AVAILABLE PRODUCTS]----
Toy Name: Teddy
Price: 100
Stock: 5

Buyer purchase:
Enter the name of the toy to buy: Teddy

Successfully bought Teddy for $100
Remaining Balance: $900

If not enough balance:
Not enough balance to buy this toy

If out of stock:
Toy is out of stock

REQUIREMENTS

- Python 3.x
- Works in: Terminal / Command Prompt, VS Code / IDLE / PyCharm

PROBLEM STATEMENT :

Running a small toy shop involves a number of things: keeping track of what stock you have, recording the quantities, pricing the goods, recording sales and keeping the books. If you're doing all of this manually then it's going to be a very time consuming job and there's a real risk of recording errors, of having an incorrect stock take and of not being able to tell what the shop's sales are. Without proper records and regular reports it's very difficult for a seller to work out what products are selling and how much the shop is actually making.

Research into computerised inventory management and accounting systems will go some way towards establishing the value of automating these functions. For this project we'll be drawing on a range of relevant literature and official sources on inventory control, point of sale and computerised business records to underpin the need for a more systematic and efficient approach.

To deal with these issues the project is to develop a very simple command line based Toy Shop Management System in Python. The system will enable the seller to add products, update prices and stock levels and view both the inventory and sales reports. The buyer will be able to view the toys available and then purchase a product against a

pre-defined balance. Following each purchase the system will update the stock levels, reduce the buyer's balance by the amount of the purchase and record the transaction for inclusion in the sales report.

The system being proposed is for basic stock and transaction management in a small toy shop. It is not going to be a full e-commerce solution like Shopee or Lazada (no online payments, no delivery, no customer accounts, no online product listings etc). It's simply going to show how a fairly simple Python script can be used to assist with a few of the shop management tasks.

PROBLEM OBJECTIVES :

A. **General Objective -** To develop a simple Python-based Toy Shop Management System that manages toy inventory, processes basic purchasing transactions, tracks the buyer's balance, and generates inventory and sales reports.

B. **Specific Objectives**

- To allow users to choose between Seller and Buyer roles.
- To allow sellers to add new toys to the inventory.
- To allow sellers to update the prices and stock quantities of existing toys.
- To display the total number of toys currently in stock.
- To allow buyers to view available toys, their prices, and their stock quantities.
- To check whether the buyer has enough balance to purchase a toy.
- To automatically decrease the toy's stock quantity after a successful purchase.
- To automatically deduct the purchase price from the buyer's balance.
- To record successful purchasing transactions for sales monitoring.
- To generate inventory reports showing product names, prices, stock quantities, and total stock.
- To generate sales reports showing the products sold, quantities sold, individual prices, and total sales revenue.
- To provide error messages for invalid inputs and unavailable products.
- To allow users to exit the program safely.

PROBLEM FEATURES :

1. **Role Selection** - Users can choose to become a Seller, Buyer, or Exit the program.
2. **Toy Inventory** - Stores toy names, prices, and quantities using a dictionary.
3. **Add Product** - Sellers can add a new toy to the inventory.
4. **Update Product** - Sellers can update an existing toy's price and stock.

5. **Stock Counter** - Displays the total number of toys currently available.
6. **Product Display** - Buyers can see the available toys, prices, and stock
7. **Balance System** - The buyer starts with a set amount of money.
8. **Purchase System** - Buyers can purchase toys if they have enough money and the product is in stock.
9. **Stock Deduction** - Stock decreases by one after a successful purchase.
10. **Balance Deduction** - The toy's price is deducted from the buyer's balance.
11. **Input Validation** - Prevents invalid numerical inputs and invalid menu choices.
12. **Exit System** - Allows the user to safely exit the program.
13. Inventory Reporting - Generates a report displaying the names, prices, and remaining stock quantities of all products, along with the total stock available.
14. Sales Reporting - Generates a report of successful transactions, including the products sold, quantities sold, individual prices, and total sales revenue. The report helps the seller monitor sales performance and review transaction records.

PLANNED INPUTS AND OUTPUTS :

A. **INPUTS** - The program will receive the following inputs from the user :

I. Role Selection

A → Seller

B → Buyer

C → Exit

II. Seller inputs

- Toy Name
- Toy Price
- Toy Quantity
- Menu Choices

III. Buyer inputs

- Selected toy name
- Menu choices

IV. Reporting Inputs

- Seller's choice to view an inventory report or sales report

B. **OUTPUTS -** The Program will display :

- Seller and Buyer menus
- Current inventory
- Total stock
- Toy names, prices, and quantities
- Buyer's current balance
- Successful purchase messages
- Remaining balance after purchasing
- Error messages for invalid inputs
- Messages for unavailable or out-of-stock toys
- Exit message
- Inventory report containing product names, prices, and remaining stock quantities
- Total number of toys currently in stock
- Sales report containing recorded transactions, products sold, quantities sold, and individual prices
- Total sales revenue from successful purchases

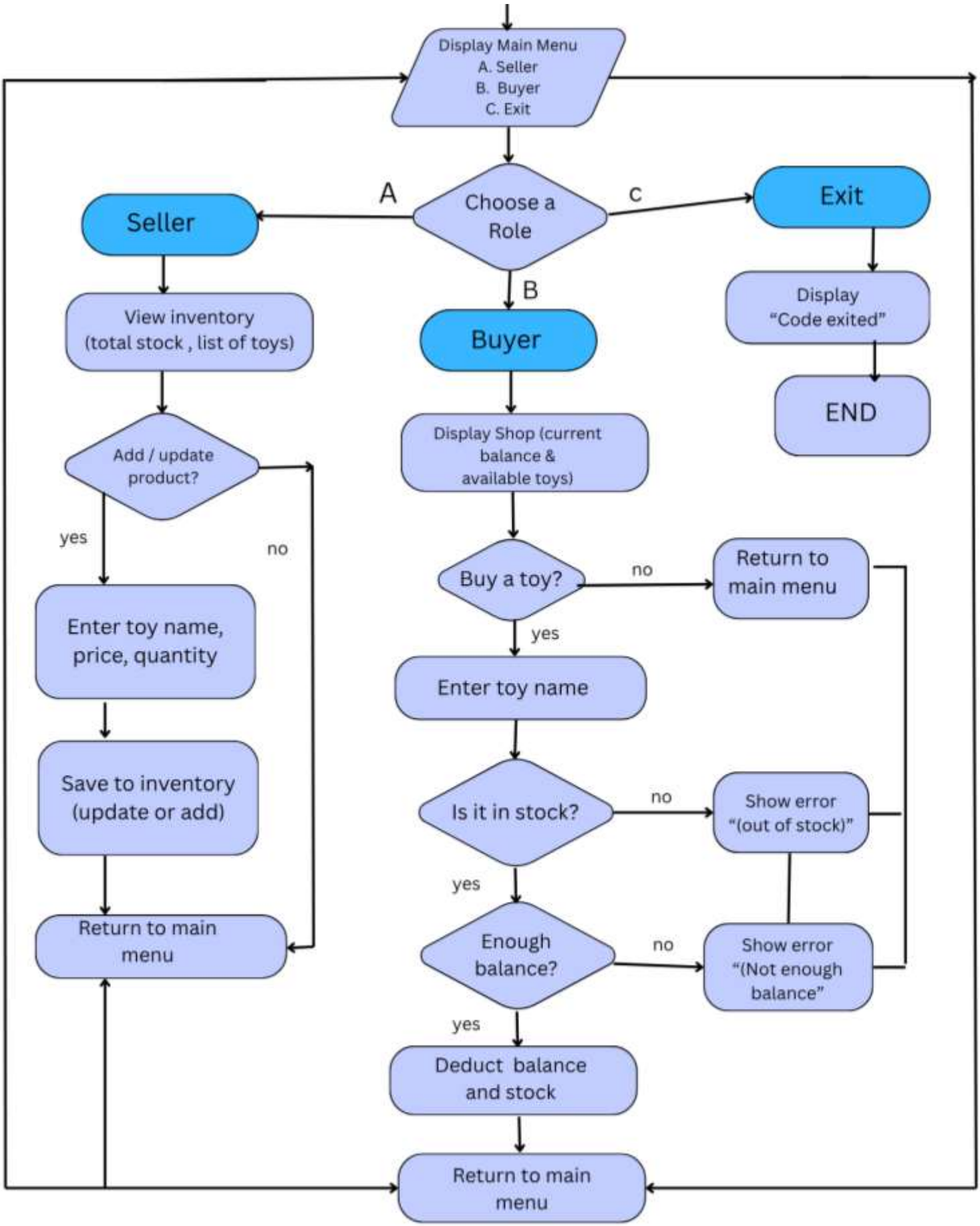
SOURCES

Shabani, A., Maroti, G., de Leeuw, S., & Dullaert, W. (2021). Inventory record inaccuracy and store-level performance. International Journal of Production Economics, 235, Article 108111. https://doi.org/10.1016/j.ijpe.2021.108111

Usita, M. M., Escultor, M. D., Garcia, A. B., Manuel, A. R., & Guinanao, G. O. (2025). Evaluating the acceptance of iProSES: A web based inventory system for food sector MSMEs using the Technology Acceptance Model (TAM). Asian Journal of Economics, Business and Accounting, 25(7), 43–54. https://doi.org/10.9734/ajeba/2025/v25i7187

(Shabani et al. (2021) looked at inventory record accuracy and retail performance using data from 81 stores and found that inaccurate records can lead to stockouts and lost revenue. Usita et al. (2025) looked at a web-based inventory management system for

small and medium sized food businesses in the Philippines and found it to be a success in terms of inventory accuracy and operational efficiency. Both studies point to the value of good inventory records and the potential advantages of utilising a computerised system to help with inventory management.)


Display Main Menu
A. Seller
B. Buyer
C. Exit
Choose a Role
A
B
C
Seller
Buyer
Exit
View inventory (total stock , list of toys)
Add / update product?
yes
no
Enter toy name, price, quantity
Save to inventory (update or add)
Return to main menu
Display Shop (current balance & available toys)
Buy a toy?
no
Return to main menu
yes
Enter toy name
Is it in stock?
no
Show error "(out of stock)"
yes
Enough balance?
no
Show error "(Not enough balance"
yes
Deduct balance and stock
Return to main menu
Display "Code exited"
END